Embedded Computer Vision Compression Techniques for Soccer Analysis

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The thesis is submitted to University College Dublin in part fulfilment of the requirements for the degree of Master of Electronic and Computer Engineering.

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April 2026

Contents

1 Abstract **1**

2 Lay Abstract **2**

3 Introduction **2**

4 Problem Aims **3**
4.1 Model Training and System Creation . . . 3
4.2 Optimization . . . 4
4.3 Compression . . . 4

5 Background Literature Review **4**
5.1 Literature Review: Overview . . . 5
5.2 Literature Review: Common Approaches . . . 5
5.2.1 Common Approaches: Multiple Object Tracking . . . 6
5.2.2 Common Approaches: Key-point Detection . . . 7
5.3 Literature Review: Current Limitations . . . 8
5.4 Modern YOLO Architecture . . . 9
5.4.1 YOLO11 Architecture . . . 11
5.4.2 YOLO26 Architecture . . . 11
5.5 Literature Review: Compression Techniques . . . 12
5.5.1 Efficiency Techniques: Hardware Acceleration . . . 12
5.5.2 Compression Techniques: Quantization . . . 14
5.5.3 Compression Techniques: Pruning . . . 14
5.5.4 Compression Techniques: Input Resolution Scaling . . . 15
5.5.5 Compression Techniques: Knowledge Distillation . . . 16

6 Methodology **17**
6.1 Model Training Procedure . . . 17
6.2 Compression . . . 18
6.3 Hardware Acceleration . . . 18
6.3.1 Quantisation . . . 19
6.3.2 Pruning . . . 20
6.3.3 Knowledge Distillation . . . 20
6.4 Evaluation Metrics and Benchmarking Protocol . . . 21
6.5 Post-Processing and Higher-level Statistic Generation . . . 22
6.6 Full-Stack System Overview . . . 23

7 Results **23**
7.1 Selecting Model Architecture and Dataset . . . 23
7.2 Summary of Model Training . . . 24
7.3 Compression Analysis . . . 25
7.3.1 Hardware Acceleration . . . 28
7.3.2 Quantisation . . . 29
7.3.3 Pruning . . . 30
7.3.4 Knowledge Distillation . . . 31
7.3.5 Combinations of Different Techniques . . . 32
7.4 Hardware Analysis . . . 33

7.4.1 Latency . . . 33
7.4.2 GPU Usage . . . 33
7.4.3 CPU Usage . . . 34
7.4.4 Power Draw . . . 35
7.4.5 Temperature . . . 36
7.5 Full-Stack System . . . 36
7.6 Full-Stack System Implementation . . . 36
7.6.1 Full-Stack System Performance . . . 36

8 Discussions and Conclusions **40**
8.1 Discussion of Results . . . 40
8.2 Limitations . . . 42
8.3 Conclusion . . . 43

9 Impact of Research **43**
9.1 Potential Beneficiaries and Pathways to Impact . . . 44
9.2 Ethical Considerations and Risk Mitigation . . . 45
9.3 Sustainability Considerations and Sustainable Development Goals . . . 46

10 Suggestions for Future Work **47**
10.1 Extending Full-Stack Match Understanding . . . 47
10.2 System-Level Optimisation for Real-Time Deployment . . . 48
10.3 Advanced Compression and Training Strategies . . . 48
10.4 Robustness, Generalisation, and Cross-Platform Validation . . . 49
10.5 Research Value of Continued Development . . . 50

11 appendices **57**

1 Abstract

This thesis investigates how embedded computer vision models for soccer analysis can be trained, compressed, and deployed efficiently on edge hardware. The work focuses on object detection and pitch keypoint localisation using YOLO11 and YOLO26 model families, with benchmarking carried out on an NVIDIA Jetson Orin Nano Dev Kit. After an initial architecture and dataset selection study, a full experimental pipeline was developed covering baseline training, hardware acceleration, quantisation, pruning, knowledge distillation, and combinations of these techniques. In total, 488 model artefacts were generated and evaluated across the object detection and pose estimation tasks.

The results show that hardware acceleration preserves analytical performance almost exactly while delivering substantial latency reductions, with average improvements of 31.3% for object detection and 29.9% for pose estimation. Quantisation produced the largest deployment gains, reducing latency by an average of 56.6% for object models and 59.6% for pose models, while also giving the strongest reduction in power usage. However, these benefits came with a larger accuracy cost, particularly for pose estimation. Pruning provided moderate latency improvements for object detection but caused much larger accuracy degradation in the pose task, indicating that the keypoint models were less robust to parameter removal. Knowledge distillation on its own had only a minor effect on latency, but remained useful as a supporting technique for preserving accuracy in smaller deployable models.

These compressed models were then integrated into a full-stack soccer analysis system for player and ball detection, pitch localisation, team assignment, possession estimation, and pass detection. Under the strongest replay setting, the best full-stack configuration achieved a Pass $mAP_{1\text{–}5}$ of 0.697 and 39.61% GPU utilisation, while reaching an effective throughput of 13.02 FPS on the target device. Overall, the thesis shows that deployment-focused compression can substantially improve the practicality of embedded soccer analytics while retaining useful analytical performance, and that a full computer vision pipeline for match statistic generation can be run effectively on constrained edge hardware.

2 Lay Abstract

This thesis looks at how soccer video analysis can be run on a small, low-power computer instead of relying on expensive cloud or server hardware. The work trained many versions of software that can find players, the ball, and the pitch in match footage, and then tested different ways of making that software smaller, faster, and cheaper to run. These included speeding up the software for the target device, reducing numerical precision, removing less useful parts of the models, and using larger models to help train smaller ones.

The study found that some of these methods gave large speed and power improvements while still keeping the analysis accurate enough to remain useful. The most effective overall approach was to combine efficient deployment methods with carefully chosen compressed models, rather than applying the most aggressive simplifications everywhere.

The final system was able to combine these models into a complete match-analysis pipeline that estimates team possession and detects passes from broadcast video. The best version of the full system reached an effective speed of about 13 frames per second on the target edge device, while keeping GPU usage below 40%. This shows that practical, on-device soccer analytics is possible on compact hardware, which could help smaller teams, clubs, and broadcasters gain access to lower-cost performance analysis tools.

3 Introduction

The use of advanced analytical methods in sport gained popularity in the mid-1990s through sabermetric principles in professional baseball [12]. Since then, data-driven analysis has become central to performance evaluation, tactical decision-making, and modern sports broadcasting. During live coverage, high-level statistics such as possession, shot counts, and pass completion rates are commonly shown alongside the scoreline and elapsed time. Historically, however, the collection of this data has typically been outsourced to specialised and expensive third-party analytics providers. For example, major English and Scottish football leagues have long-term data rights agreements with Opta, granting exclusive responsibility for the collection and distribution of official match statistics across

broadcast, media, and analytical platforms [36].

More recently, the growth of online streaming platforms has increased the broadcast coverage of lower-budget and regional sporting events [27]. At the same time, amateur and semi-professional teams are increasingly adopting data-driven approaches to support training and performance evaluation [11]. Advances in computer vision now make it possible to analyse sports video streams in real time, enabling low-cost, on-site generation of match statistics without reliance on cloud infrastructure. This could reduce costs for both sports broadcasting and sports science applications, with particular benefits for lower-budget organisations.

These trends motivate the aim of this work: **the optimisation and compression analysis of embedded computer vision models for real-time soccer analysis on an edge device.**

4 Problem Aims

4.1 Model Training and System Creation

The first and most basic objective of this project is to train computer-vision models that can automatically extract low-level information from video of soccer games. These models should be trained with the intention of being run on an edge device, but should have an mAP_{50} within 10% to that of other modern, non real-time approaches [3]. Mean Average Precision (mAP) is a common metric for object detection models that measures the accuracy of predicted bounding boxes by combining precision and recall across different confidence thresholds. mAP_{50} specifically refers to the mAP computed using an Intersection over Union (IoU) threshold of 0.50, meaning a predicted bounding box is only counted as a correct detection if it overlaps the ground-truth box by at least 50%.

For this project, an NVIDIA Jetson Orin Nano Dev Kit will be used as the target edge device [24]. Alongside these models, an end-to-end system should be developed, to convert the video and model outputs to desired high-level game statistics. This will allow the

functionality of the system to be explored, as well as assess hardware and throughput performance of the edge device.

4.2 Optimization

The second objective is to optimise the end-to-end system with respect to computational efficiency, aiming to reduce CPU, GPU, and memory utilisation, device temperature and power draw without significantly affecting analytical performance. Specifically, the system should perform to within 15% mAP_{50} of modern object detectors [3] used for soccer analysis, while utilizing no more than 80% of the device's total CPU, GPU, and RAM availability. Similarly, the power draw and temperature should not exceed 80% of the device's recommended limits, and maintain roughly 20 FPS to allow for real-time analysis. NVIDIA recommends a maximum CPU and GPU temperature of 85° and a max power draw of 25W, and so these will used as the initial recommended limits [24].

4.3 Compression

The third objective is to apply model-level compression techniques directly to the trained models. As with before, compressed models should maintain an analytical performance within 15% mAP_{50} of the baseline uncompressed models, while targetting a FLOP (Floating Point Operations Per Second) reduction of 5x and a parameter size reduction of 5x. This will also indirectly lower GPU/CPU and power requirements, supporting the 80% total CPU, GPU, and RAM availability aims discussed earlier.

5 Background Literature Review

To achieve these aims, a background literature review will be performed to ensure that the proposed methodology described later utilises best modern practices and research, and targets a clear gap in research. Focus will be placed on an overview of the problem, common approaches and prevalent models, and common model compression techniques.

5.1 Literature Review: Overview

As discussed in the introduction, modern sporting preparation and broadcast have put a larger emphasis on quantitative data than opinion-based analysis, especially with the rise of next-generation tracking and sensor technologies [30]. Some of the most useful of these implementations explore tracking key statistics such as team pass completions percentages, possession, and team formation analysis through the use of object detection/tracking and pitch recognition [45, 20]. However, emphasis in this niche has been largely placed on accuracy over computational efficiency or real-time analysis, with some designed software systems requiring over 515B Floating Point Operations Per Second (FLOPS) [32], which edge devices cannot maintain.

5.2 Literature Review: Common Approaches

Modern computer vision-based approaches to estimating possession and formation often utilise object detection models coupled with object tracking algorithms to produce Multiple Object Tracking (MOT) outputs [3]. This allows identities (in the form of bounding boxes) for on-screen outfield players, goalkeepers, referees, and the ball to be used for statistic generation. From here, clustering techniques such as K-means and mean shift methods are often applied, separating detecting players into two teams, as well as determining the goal-keepers [46]. This team assignment allows detailed estimates of possession and pass completion to be determined.

Alongside object detection for players, a variety of different key-point detection strategies are often used to detect the location on screen of relevant pitch markings, such as the corners, penalty box, and midfield [8]. With this and the player bounding boxes from earlier, an optical perspective transformation known as homography estimation can be applied to calculate the co-ordinate location on the pitch of each object of interest [5]. This allows player speed, distance, and position statistics to be generated.

The reported features and statistics of some implemented software solutions are displayed below in Table 1. Note that reported FPS values are taken from original implementa-

tions, and are not directly comparable across hardware. Also note that model accuracy and size values (mAP_{50}, FLOPS, Parameters) cannot be explored here, due to significant differences in software structures (both in amount of models, model output information, and implementation output). Instead, the accuracy and size of the models used in these will be explored explicitly in subsections **Multiple Object Tracking** and **Key-point Detection** later.

Implementation	Primary Models Used	FPS	Features	Real-time?
RoboFlow [32]	YOLOv10m-pose (Pitch) YOLOv8x (Player) YOLOv8x (Ball)	< 1	Position Map	No
CodeJiffy [22]	YOLOv11n-pose (Pitch) YOLOv11s (Players, Ball)	~ 7	Position Map, Possession Speed Estimation	No
SoccerMaster [41]	QWEN2.5 (Jerseys) YOLOv8 (Players)	< 1	Event Detection, Position Map Jersey Number Detection	No
HMBZO [10]	YOLOv8m (Pitch) YOLOv8l (Players, Ball)	~ 9	Position Map	Yes

Table 1: Previously Implemented Football Analysis Software Solutions

Note: Accuracy and size comparisons (mAP_{50}, FLOPS, Parameters) are omitted due to architectural differences.

5.2.1 Common Approaches: Multiple Object Tracking

The object detection portion of the above solutions are commonly addressed using YOLO (You Only Look Once) models [28]. YOLO models utilise a single pass for both localisation and detection, resulting in a low-latency detection architecture ideal for real-time analysis. This has been previously explored in soccer analysis, with models producing mAP_{50} results of up to 87%, and mAP_{50-95} of up to 64.9% using labelled broadcast soccer videos [18]. mAP_{50-95} represents the model's mAP for all IoU thresholds between 50%-95%. Another potential real-time detection model is the Faster R-CNN model. Faster R-CNN utilises a 2-stage pipeline, which first separates frames into region proposal networks, before performing classification on a second pass [31]. While used in post-match processing, Faster R-CNN is likely not applicable to real-time scenarios, due to this heavy 2-stage inference process. Table 2 below summarizes some commonly used models for real-time object detection. Note

that latency cannot be compared directly here, as models have not been benchmarked on equal GPUs.

Model	**AP$_{50-95}$** (%)	**Parms** (M)	**FLOPS** (B)
YOLO [28, 38]			
v11-n	39.5	2.6	6.5
v11-s	47.0	9.4	21.5
v11-m	51.5	20.1	68.0
v8-n	37.3	3.2	8.7
v8-s	44.9	11.2	28.6
v10-s	46.8	N/A	21.6
RT-DETR-R18 [44]	46.5	19.2	60
R50-FPN [31, 15]	41.0	60	160

Table 2: Commonly Used Object Detection Models

A number of commonly used and publicly available datasets exist for the training of these models. The most popular and utilised datasets are displayed below.

SoccerNet Dataset [3]

Class	**Bounding Boxes**
Player	2,992,173
Goalkeeper	130,109
Referee	301,025
Ball	215,156
Total	**3,645,661**

Roboflow Dataset [33]

Class	**Bounding Boxes**
Player	13,241
Goalkeeper	473
Referee	1,519
Ball	565
Total	**15,798**

Table 3: Commonly Used Datasets in Literature

The tracking of these bounding boxes is then implemented (MOT); modern approaches often use the SORT algorithm, in which object identities are maintained using Kalman filters for motion prediction and the Hungarian algorithm for object association [1].

5.2.2 Common Approaches: Key-point Detection

As discussed earlier, on-screen locations of key pitch marking are desired to allow homography estimation to be applied. Computer vision models are commonly used for this as well. Offline systems often use encoder–decoder architectures, which estimate pitch key locations by compressing image features into a latent representation and then decoding this

representation to produce precise key-point predictions [2]. More recently, YOLO-based pose detection models have been applied [17], utilising their single pass architecture to enable low-latency key-point estimation. Figure 4 displays some models that will be initially explored. As with before, latency comparisons are not feasible due to benchmarking differences between literature.

Model	**mAP**$_{50-95}$	**Params (M)**	**FLOPS (B)**
YOLO-Pose [28, 38]			
v11n-pose	50.0	2.9	7.4
v11s-pose	58.9	9.9	23.1
v11m-pose	64.9	20.9	71.4
v8n-pose	48.7	3.3	8.9
v8s-pose	56.8	11.6	30.2
HRNet-W32 [40]	66.4	28.5	95.0
OpenPose-Light [26]	54.0	25.9	160.0

Table 4: Keypoint Detection Models

Some classical, non-neural approaches also exist, which align on-screen edges with known pitch dimensions to project player positions into real-world coordinates without learned representations [16]. While this enables significantly faster throughput and avoids the need for training data, performance is often sensitive to occlusion, inaccurate line detection, and changes in broadcast viewpoint or visual clutter, limiting robustness in real match conditions.

5.3 Literature Review: Current Limitations

Despite the rapid progress in computer vision for sports, several significant technical and operational hurdles remain unexplored for real-time edge deployment. As Table 1 displayed earlier, common models used today [10, 41, 22, 32] are computationally intensive, demanding high-end GPU resources (e.g, NVIDIA A100 TensorRT) that are typically unavailable on constrained edge devices like the Jetson Nano. Most systems are designed for batch frame processing, and require 200+ of GFLOPS to produce < 1 FPS in real-time [41, 32]. Other systems, while feasible for real time, assume much more powerful infrastructure and still produce only 5-10 FPS on consumer hardware[10, 22]. This leads to

systems and software unsuitable for edge deployment without first applying optimisation or compression.

5.4 Modern YOLO Architecture

Recent YOLO detectors generally follow a consistent *backbone–neck–head* design, where the backbone extracts increasingly abstract visual features, the neck fuses multi-scale information, and the head produces predictions for different object sizes [38]. In these modern variants, the backbone typically begins with stride-2 convolutional layers that progressively reduce spatial resolution while increasing channel depth. This is followed by repeated feature extraction blocks, which differ by version but serve the same general purpose of learning richer hierarchical representations. The neck usually combines high-level semantic information with finer spatial detail through repeated upsampling and concatenation, enabling the detector to preserve both contextual and localization information across scales. Finally, the head produces predictions at three resolutions, corresponding broadly to small, medium, and large objects [38].

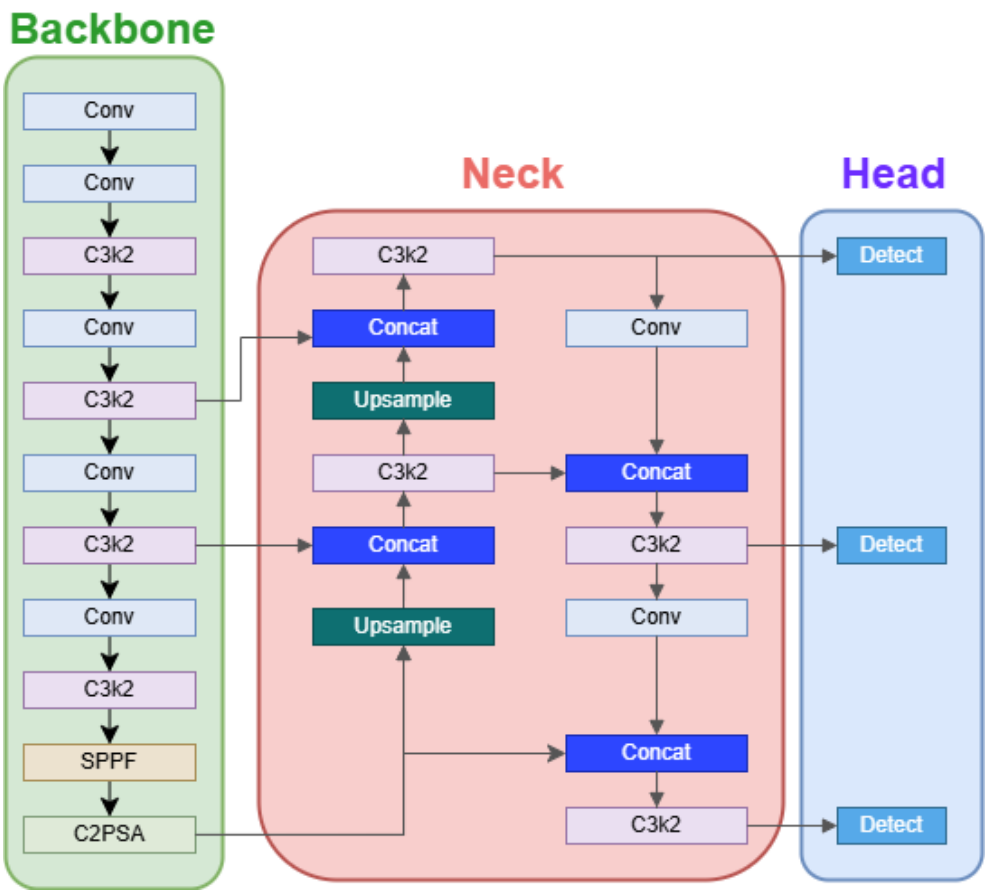


Figure 1: Simplified overview of the YOLO11 architecture, showing the backbone, neck, and head structure.

A major theme in modern YOLO development is that improvements are typically introduced through refinements to internal building blocks rather than through complete architectural redesign. Across recent versions, this has included lighter feature extraction modules, improved multi-scale pooling, and the increasing use of attention mechanisms to strengthen global context modelling. In parallel, head design has also evolved toward more efficient and deployment-oriented inference, with newer models seeking to reduce computational overhead while maintaining strong detection performance. As a result, recent YOLO models can be understood as sharing a common single-stage detection framework while differing in how they optimize feature extraction, attention, training strategy, and inference efficiency [38].

5.4.1 YOLO11 Architecture

YOLO11 retains the standard modern YOLO backbone–neck–head pipeline, but introduces several architectural refinements relative to earlier Ultralytics versions. Its backbone begins with two stride-2 convolutional layers, after which the main feature extraction stages are built around the *C3k2* block rather than the *C2f* block used in earlier models such as YOLOv8. The reviewed paper describes C3k2 as a lighter and more advanced replacement intended to reduce parameter count while preserving strong representational capacity. It also notes that the internal behaviour of C3k2 depends on variant, with larger YOLO11 models consistently using the *C3k* path. This makes YOLO11 an example of the broader trend toward improving efficiency through more carefully designed internal modules rather than major changes to the overall detector layout [38].

After the backbone, YOLO11 applies *SPPF* and then *C2PSA* before entering the main neck fusion pathway. SPPF provides multi-scale pooling, while C2PSA introduces a self-attention mechanism to improve global feature modelling. Importantly, the attention block is placed only at the lowest-resolution stage in order to limit computational cost. The neck then uses repeated upsampling and concatenation operations to merge deep semantic features with higher-resolution features from earlier backbone stages, before passing them to a three-scale detection head for small, medium, and large object detection. Overall, YOLO11 preserves the established multi-scale YOLO framework, but distinguishes itself through the use of C3k2, C2PSA, and related lightweight convolutional refinements that improve efficiency without fundamentally altering the architecture [38].

5.4.2 YOLO26 Architecture

YOLO26 is architecturally very similar to YOLO11 and preserves the same overall backbone, neck, and head structure, including the use of stride-2 convolutions in the stem, *C3k2* blocks for feature extraction, *SPPF* and *C2PSA* near the bottleneck, and a three-scale detection head [38]. However, the literature describes YOLO26 as an incremental refinement of YOLO11 rather than a direct continuation without change. One architectural modification is the addition of a shortcut connection inside the *SPPF* block, which is

intended to improve information flow and stabilize optimization at high semantic levels. A second change is in the final *C3k2* block before detection, where YOLO26 incorporates an attention mechanism through a *PSABlock*, aiming to strengthen global context modelling without a large increase in parameters or latency [38].

The more substantial differences between YOLO26 and YOLO11 occur in the detection and training pipeline. The paper states that YOLO26 removes *Distribution Focal Loss (DFL)* from the detect block and instead uses direct box regression, simplifying both training and inference. It also adopts a dual-assignment, end-to-end, NMS-free design inspired by YOLOv10, where one-to-many and one-to-one assignments are both used during training, but only the one-to-one path is retained at inference. In addition, YOLO26 introduces *Progressive Loss Balancing (ProgLoss)* and *Small-Target-Aware Label Assignment (STAL)* to improve training stability and enhance learning for small objects, alongside the *MuSGD* optimizer for smoother convergence. Consequently, while YOLO26 looks very similar to YOLO11 at the high level, it differs in several important architectural and training details that push it further toward edge-oriented, end-to-end, and small-object-aware detection [38].

5.5 Literature Review: Compression Techniques

A range of model compression techniques designed to reduce the computational and memory requirements of deep learning models (while preserving acceptable analytical performance) have been explored in literature. Although these techniques have not yet been widely applied to end-to-end football statistic generation systems, they will form a strong foundation for deploying real-time soccer analysis pipelines on edge devices.

5.5.1 Efficiency Techniques: Hardware Acceleration

Hardware acceleration improves inference efficiency by executing neural networks through runtimes and software stacks that are specifically optimized for the target hardware platform [47]. Rather than relying on a default framework execution path, accelerated backends can apply graph-level optimizations, fuse operations, select more efficient kernels, and bet-

ter exploit GPU parallelism. This is particularly important in real-time computer vision, where inference latency directly affects whether a system can operate at interactive or live processing speeds.

A common approach to hardware acceleration in embedded and GPU-based deep learning systems is the use of optimized inference runtimes such as TensorRT. These frameworks convert trained models into deployment-ready representations that are tailored to the hardware on which inference will be performed. By moving much of the optimization work offline, such approaches reduce runtime overhead and improve throughput, making them especially valuable in edge settings where compute and power budgets are constrained.

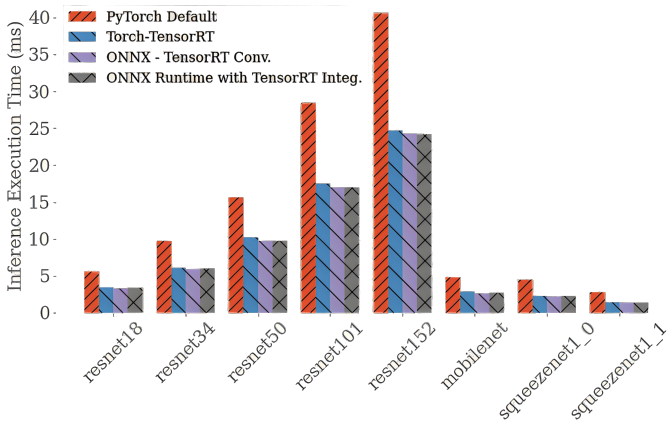


Figure 2: Effect of `tensorrt` application on inference time.

Figure 2 illustrates the typical effect of hardware acceleration on inference execution time [47]. As can be seen, optimized TensorRT-based pipelines consistently reduce latency relative to the default PyTorch implementation, with similar trends also observed for ONNX-based accelerated deployment. This demonstrates that hardware acceleration can provide substantial efficiency improvements without altering the underlying model task, making it an important optimization strategy in the deployment of real-time computer vision systems.

5.5.2 Compression Techniques: Quantization

Quantization is one of the most widely adopted compression techniques, and involves reducing the numerical precision of model parameters and activations [7]. While standard, uncompressed neural networks typically utilise 32-bit floating-point (FP32) representations, quantization maps these values to lower-precision formats such as 8-bit (INT8) or 4-bit (INT4) integers. This process substantially reduces memory bandwidth usage and arithmetic complexity, with INT8 addition being up to 30× more energy-efficient and 116× more area-efficient than its FP32 counterpart on 45nm hardware [7]. Figure 3 below displays the effect of different quantization levels on device power consumption.

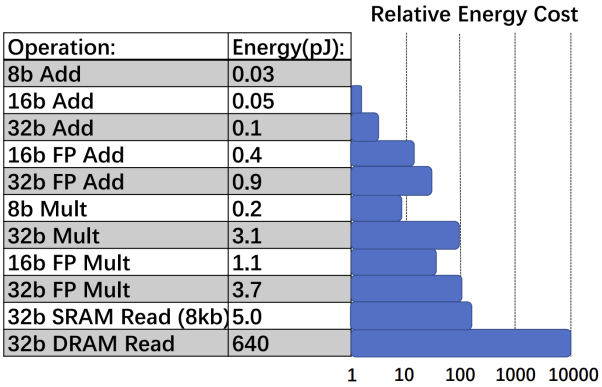


Figure 3: Effect of Quantization of Model Energy Consumption [7]

As can be seen, identical operations utilise significantly less energy (in Joules) when using less precise values. For example, an 8-bit multiplication requires 0.2 pJ, while a 32b multiplication requires 3.1 pJ. This demonstrates the clear optimization benefit that quantization can supply.

5.5.3 Compression Techniques: Pruning

The technique of pruning aims to reduce model complexity by removing parameters that contribute minimally to the network's output [7, 42]. This technique follows the assumption that the smallest system capable of fitting the training data typically yields the best generalization on unseen inputs [29]. It is often performed after training, where a loss function is used to estimate the importance of weights, before identifying and eliminating redundant connections.

Pruning strategies are fundamentally divided into unstructured and structured approaches. Unstructured pruning removes individual weights with low importance regardless of their location [7], which achieves high compression ratios at the cost of efficient memory manipulation. Structured pruning removes whole structures such as entire filters, channels, or layers, which maintains efficient memory at the cost of less compression. [42]. Figure 4 below displays model accuracy (TOP-1) and FLOPS, when various pruning strategies are applied.

	Method	Top-1	FLOPs	Top-1 ↓	FLOPs ↓ (%)
Swin-T	Baseline	81.2	4.5	0.0	0.0
	STEP	77.2	3.5	4.0	22.2
	ViT-Slim	80.7	3.4	0.5	24.4
	X-Pruner (Ours)	**80.7**	**3.2**	**0.5**	**28.9**
Swin-S	Baseline	83.2	8.7	0.0	0.0
	STEP	79.6	6.3	3.6	27.6
	WDPruning	81.8	6.3	1.4	27.6
	X-Pruner (Ours)	**82.0**	**6.0**	**1.2**	**31.0**

Figure 4: Effect of Pruning on Accuracy and FLOPS [42]

As can be seen, applying the X-Pruner algorithm reduces the required FLOPS by 28%, with only a .5% reduction in accuracy (Top-1). For this reason, pruning will be explored as a method of compressing training soccer computer vision models.

5.5.4 Compression Techniques: Input Resolution Scaling

Input resolution scaling reduces computational and memory footprints by lowering the spatial dimensions of the images or tokens processed by a model. Since the complexity of Convolutional Neural Networks and Vision Transformers typically scales quadratically with input resolution, even modest downsizing can yield substantial improvements in inference throughput and energy efficiency [21]. Figure 5 displays model characteristics (FLOPS and memory usage) before and after input resolution scaling is applied.

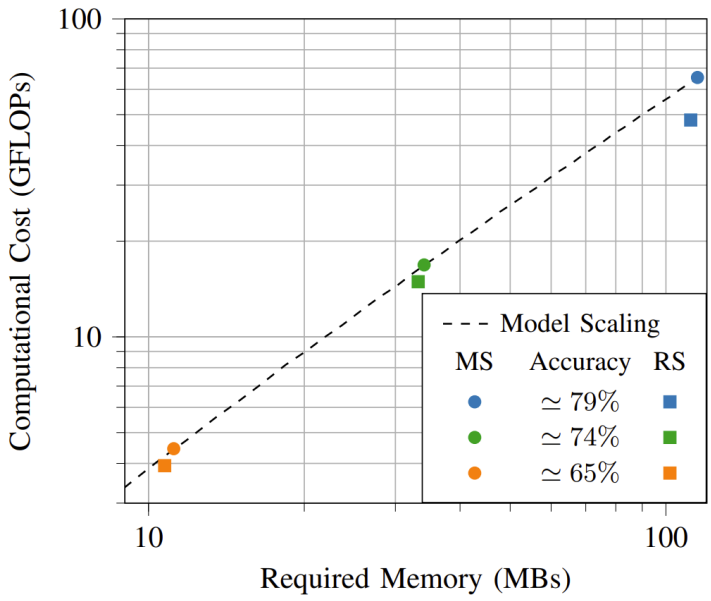


Figure 5: Required Memory Vs. FLOPS when Input Resolution Scaling is Applied [21]

As can be seen by Figure 5, applying input resolution scaling often reduces the required FLOPS by ∼2x, with slight improvements in required memory. However, this technique may introduce a major accuracy/efficiency trade-off, as reducing resolution compromises the detection of small objects [21, 45]. In the soccer domain, where the ball often occupies fewer than 100 pixels, resolution is a primary determinant of performance; for instance, increasing resolution from 640px to 1200px has been shown to improve ball detection recall by up to 96% [18]. To mitigate these losses, implementations often utilise training-evaluation discrepancies, training models on smaller resized crops but evaluating at slightly higher resolutions to recover analytical performance [21].

5.5.5 Compression Techniques: Knowledge Distillation

Knowledge distillation is another model compression technique, in which a compact "student" model is trained to replicate the performance of a high-capacity "teacher" model. Unlike standard training, the student is optimised using softened output distributions (logits) or intermediate feature representations produced by the teacher, which contain richer information than hard ground-truth labels [9, 7]. Knowledge transfer can be categorised into three types: response-based, which focuses on the final output probabilities; feature-based, which aims to mimic the teacher's internal feature maps to capture implicit

patterns; and relation-based, which teaches the student the relationships between different data samples or layers [9].

6 Methodology

The methodology will be focussed into three parts; the training of initial models, the compression of these models, and the creation and optimisation of a full-stack system to utilise these models to apply high-level analysis of broadcast soccer film.

6.1 Model Training Procedure

To support fair comparison, all uncompressed object detection and keypoint detection models were trained under a standard procedure. Training was initialised from COCO-pretrained weights [14], and then fine-tuned on a football-targeted dataset. All models were trained in PyTorch and saved as `.pt` checkpoints prior to compression. Training used an input resolution of 1280×1280, a batch size of 32, and was conducted on the UCD high-performance cluster *UCDSonic*. Optimisation used stochastic gradient descent with an initial learning rate of 0.01, momentum 0.937, and weight decay 5×10^{-4} [39]. Training ran for up to 500 epochs, with a 3-epoch warm-up schedule and early stopping patience of 60 epochs.

An initial benchmarking study was performed to identify the most suitable model family for the remainder of the work, based on analytical performance and hardware usage. A separate preliminary study was then conducted to determine the most suitable dataset for the full object detection study, in which 10 model variants were trained under identical conditions and evaluated across three labelled football broadcast datasets: SoccerNet [3], Roboflow [32], and Football Analytics [34]. For pitch keypoint detection, the Roboflow football field dataset [35] was used throughout.

6.2 Compression

As discussed in the Literature Review, three chief compression techniques have been explored: quantisation, knowledge distillation, and pruning. Figure 6 below displays the various combinations of compression techniques applied.

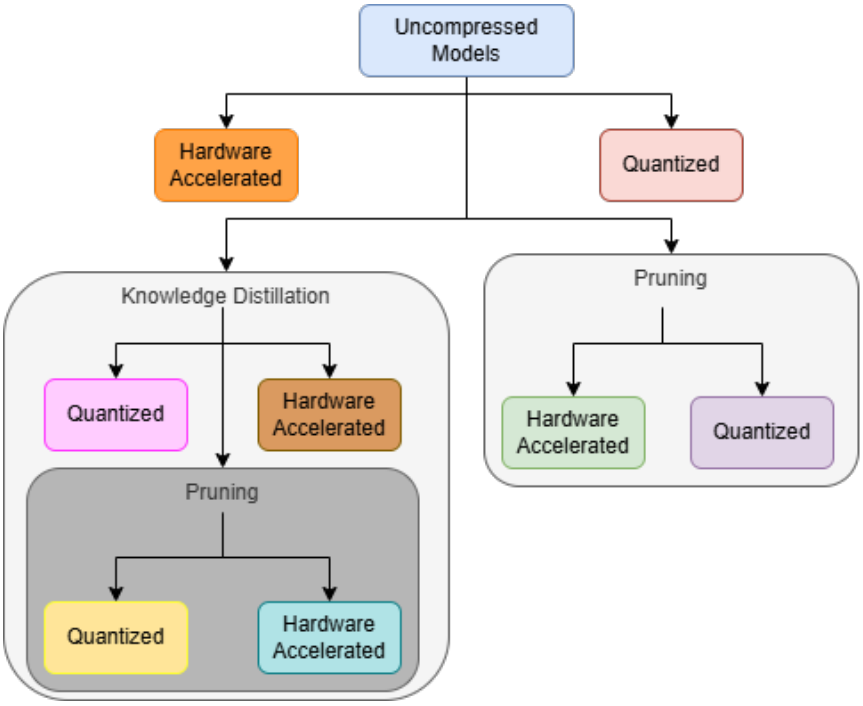


Figure 6: Proposed full-stack system diagram.

In total, 11 compression configurations are applied to the baseline uncompressed models. These comprise the four individual techniques—hardware acceleration, quantisation, knowledge distillation, and pruning—together with seven hybrid configurations formed by combining these methods in different ways.

6.3 Hardware Acceleration

After training in PyTorch, each model checkpoint was exported from its native `.pt` format to a TensorRT `.engine` file in FP32 precision to enable hardware-accelerated inference on the NVIDIA Jetson Orin Nano. This export stage was included so that evaluation could be performed using the deployment runtime intended for the target edge device, rather than relying on the original training framework. While FP32 retains full numerical precision, conversion to TensorRT still enables substantial optimisation through techniques such as

graph simplification, layer fusion, and improved kernel scheduling. This made FP32 engine export a suitable deployment format for measuring model performance under realistic edge inference conditions while avoiding changes to the underlying numerical representation.

This approach was used because it reduces inference overhead and improves deployment efficiency on embedded GPU platforms. NVIDIA recommends separating engine building from inference itself: the engine is built and serialized once, then loaded later for repeated inference runs. This avoids repeating the expensive optimization stage each time the application starts and allows TensorRT to exploit hardware-specific optimizations such as reduced-precision execution. A further practical consideration is that .engine files are generally tied to the TensorRT version and hardware platform on which they were created, meaning that engines usually need to be rebuilt if the deployment environment changes significantly.

6.3.1 Quantisation

FP16 and INT8 quantisation was implemented through post-training export of trained PyTorch checkpoints to TensorRT `.engine` files, enabling reduced-precision inference on the NVIDIA Jetson Orin Nano. Rather than modifying the model training stage, quantisation was applied at deployment time in order to reduce computational and memory requirements on the target edge device. All exports were performed at a fixed input resolution of 1280×1280 and with ONNX opset 20 to maintain consistency across models [7, 23]. This allowed quantisation to be evaluated as a practical deployment-stage compression technique for improving inference efficiency while preserving acceptable model performance.

For INT8 engine generation, calibration data was provided during export to estimate representative activation ranges before deployment [23]. Task-specific dataset definitions were used for object detection and pose estimation, with calibration samples drawn from the training split by default and optionally limited by dataset fraction. The resulting TensorRT engines were then evaluated through the same benchmarking and validation pipeline as the other deployed models, allowing their impact on latency, accuracy, and hardware usage to be assessed consistently on the Jetson Orin Nano.

6.3.2 Pruning

Pruning was implemented as structured channel pruning using NVIDIA ModelOpt Fast-NAS [29, 23]. This approach searches for a reduced channel configuration under an explicit FLOP constraint, producing smaller and more efficient network variants while preserving a deployment-compatible architecture. In this work, trained YOLO checkpoints were pruned using target FLOP reductions of 90%, 80%, and 70% where feasible. After this, fine-tune training was applied under the same procedure as the uncompressed models, but for only 100 epochs.

6.3.3 Knowledge Distillation

Knowledge distillation was implemented as a teacher–student training strategy, in which a smaller YOLO student model was optimised using both the standard detection loss and an additional distillation loss. A larger pretrained YOLO teacher model was kept fixed throughout training, with no gradient updates, and both models processed the same input image at each iteration. Rather than distilling only the final predictions, supervision was applied at intermediate feature levels by matching corresponding stages in the teacher and student backbone and neck. Since these paired feature maps did not always have the same channel dimensions, lightweight alignment layers were used to project the student features into the teacher feature space before the distillation loss was computed. This allowed the student to retain its lightweight architecture while benefiting from the stronger internal representations learned by the teacher.

Two feature-distillation methods were evaluated: Channel-Wise Distillation (CWD) and Masked Generative Distillation (MGD) [9, 37]. CWD encourages the student to match the teacher's channel-wise spatial response patterns, helping it learn more discriminative feature representations. MGD instead applies random masking to the student features and trains a small reconstruction module to recover teacher-like representations, promoting stronger semantic understanding and robustness. The overall training objective was defined as

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_t \mathcal{L}_{\text{distill}},$$

where λ_t followed a decaying schedule, with a larger value early in training and a smaller value later. This placed greater emphasis on teacher guidance during the early stages of representation learning, before gradually shifting focus toward the student's own detection performance.

6.4 Evaluation Metrics and Benchmarking Protocol

Model accuracy was evaluated primarily using $\text{mAP}_{50\text{–}95}$ on a validation subset [6]. $\text{mAP}_{50\text{–}95}$ is a standard object detection metric that measures how well predicted objects match the ground truth in both classification and localisation, describing the detection performance averaged across Intersection over Union (IoU) thresholds from 0.50 to 0.95 in steps of 0.05. This makes it a stricter measure than evaluating at a single threshold. It is expressed mathematically as:

$$\text{mAP}_{50\text{–}95} = \frac{1}{10} \sum_{k=0}^{9} \text{AP}_{0.50+0.05k}$$

where $\text{AP}_{0.50+0.05k}$ denotes the average precision computed at an IoU threshold of $0.50 + 0.05k$. Average precision (AP) is defined as the area under the precision–recall curve, summarising the trade-off between detection precision and recall for a given IoU threshold.

The metric $\text{mAP}_{1\text{–}5}$ was used to analyse pass-detection accuracy, where predictions were matched to ground-truth pass events using temporal tolerances of 1 to 5 seconds rather than spatial IoU thresholds. This allows pass detection performance to be assessed under increasing levels of timing tolerance. Hardware usage was assessed primarily by GPU utilisation, power draw, and device temperature. These metrics were chosen for their relevance to the NVIDIA Jetson Orin Nano, which operates within an 85°C thermal limit, a 25 W power budget, and a constrained 1024-core CUDA GPU, which provides limited compute resources, especially for multiple heavy computer vision workloads run in parallel [24].

Hardware usage and inference performance were measured for each model by running inference on the same image set over a 600s run. Each run began with five warm-up inferences,

which were excluded from analysis. Latency was recorded for every timed inference, while throughput (FPS) was computed as the total number of successful inferences divided by cumulative inference time. In parallel, system telemetry was sampled every 1000ms using NVIDIA's built-in `tegrastats` command, from which mean CPU utilisation, GPU utilisation, RAM usage, temperature, and power draw were computed.

After all models were benchmarked and validated, the efficiency-accuracy trade-off could be analysed. To identify the optimal models in this regard, Pareto analysis was applied in the two-dimensional space, maximising $\text{mAP}_{50\text{–}95}$ while minimising latency [19]. A model i was considered Pareto-dominated and dismissed if there existed another model j such that:

$$\text{mAP}^{(j)}_{50\text{–}95} \geq \text{mAP}^{(i)}_{50\text{–}95} \quad \text{and} \quad \text{LAT}^{(j)} < \text{LAT}^{(i)}$$

The remaining non-dominated models form the Pareto frontier and were used to shortlist best models for full-stack analysis.

6.5 Post-Processing and Higher-level Statistic Generation

The trained and compressed object and keypoint models were then used in a post-processing pipeline to allow for higher-level football analytics to be generated. This stage converted raw model outputs into more complex match statistics: specifically team identity, player possession, team possession, and pass events. To support this, the 2D pitch-coordinate locations of players and the ball were estimated using a homography computed with RANSAC [13] and temporally stabilised with exponential smoothing. For this projection, the bottom-centre point of each player bounding box, corresponding approximately to the player's feet, and the ball centre were used. Object detections were associated across frames using the ByteTrack MOT algorithm [43].

Team identity was assigned from upper-body player crops with pitch pixels suppressed. The dominant jersey colour was estimated using 2-cluster K-means clustering [46], and the resulting dominant colour was then matched, after temporal smoothing, to manually predefined team colour palettes.

Player possession was estimated from the projected locations of players and the ball, with possession assigned to the nearest player within a predefined distance threshold. Passes were defined as valid same-team transitions in control between different players.

6.6 Full-Stack System Overview

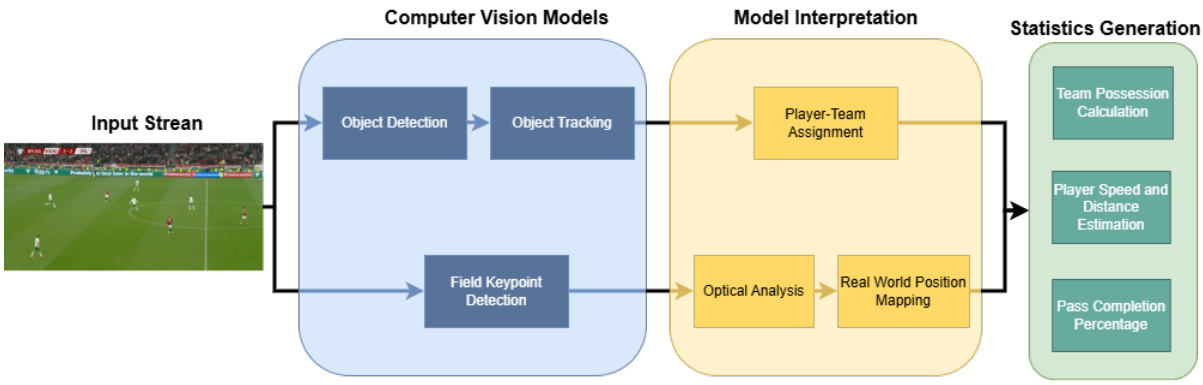


Figure 7: Proposed full-stack system diagram.

Figure 7 illustrates how the trained object detection models, keypoint localisation models, and post-inference processing were integrated into the proposed full-stack system. Each video frame was processed both by an object detector and a pitch keypoint detector, with these intermediate outputs then consumed by the post-inference algorithms described previously. Inference could be performed either on every frame or at a configurable interval, to allow further hardware efficiency customisation. Confidence thresholds were applied to object detection, pitch keypoint detection, team-colour assignment, and pass classification.

7 Results

7.1 Selecting Model Architecture and Dataset

As discussed earlier, preliminary experiments were first carried out to determine the model family and training dataset to be used for the majority of the study. These experiments compared candidate object detection architectures in terms of analytical performance and latency, and also compared several labelled football broadcast datasets.

Figure 8: Architecture and dataset comparison.

(a) Architectures. (b) Datasets.

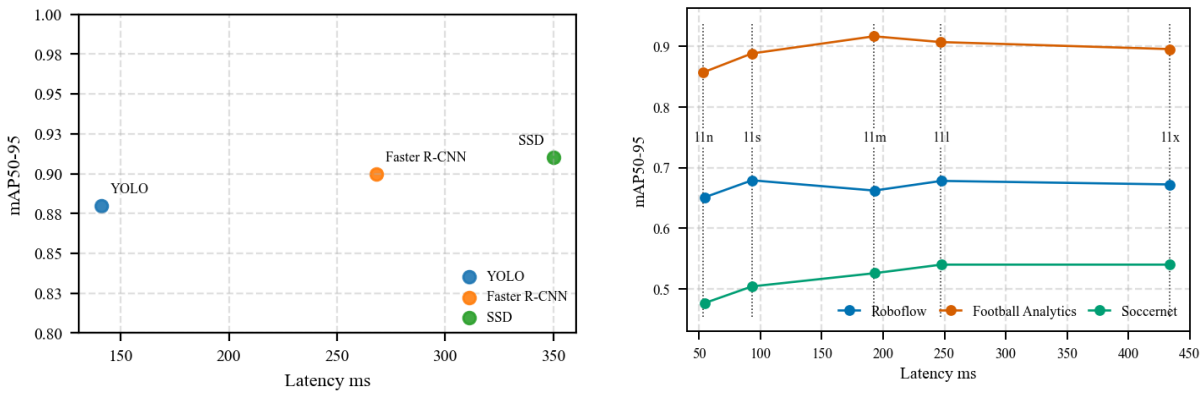


Figure 8a compares the analytical performance and latency of a range of object detection model families. Although Faster R-CNN [44] and SSD achieved slightly higher initial accuracy, the YOLO family [38] provided substantially lower latency on the Jetson Orin Nano Dev Kit. As a result, YOLO-based models were selected for the subsequent object detection experiments, as they offered the most suitable balance between accuracy and real-time edge deployment performance.

Figure 8b shows the preliminary dataset comparison carried out using the SoccerNet [3], Roboflow [32], and Football Analytics [34] datasets. Among these, the Football Analytics dataset achieved the strongest initial mAP_{50-95}, at approximately 91%, and was therefore selected for the majority of the object detection study.

7.2 Summary of Model Training

With the model family and dataset selected, the full training and compression study was then carried out. In total, 488 unique model artefacts were generated and evaluated across the object detection and pitch keypoint tasks. This included 104 **YOLO11** models, 152 **YOLO11-pose** models, 72 **YOLO26** models, and 160 **YOLO26-pose** models. Table 5 summarises the final model inventory used throughout the study.

Table 5: Model inventory.

Category	**Count**
Total Models	488
Object Models	212
Pitch Models	276
Uncompressed	40
Accelerated	40
Distilled	70
Pruned	68
Quantized	40
Combinations	270

As shown in Table 5, the majority of the model inventory consisted of compressed variants rather than baseline models, reflecting the central aim of the study: to examine how different deployment-focused compression strategies affect performance on edge hardware.

7.3 Compression Analysis

This section analyses the effect of the main optimisation and compression strategies explored in this work, namely hardware acceleration, quantisation, pruning, knowledge distillation, and combinations of these techniques. For each approach, the change in mAP50-95 and latency is examined in order to assess how deployment efficiency can be improved while maintaining acceptable model accuracy. Together, these subsections provide a structured comparison of the main accuracy–efficiency trade-offs, and help identify the compression pathways that are most suitable for real-time embedded soccer analysis.

Tables 6 and 8 condense the benchmark results into task-specific averages for the base techniques, while Tables 7 and 9 isolate the combined compression paths. Each value is measured relative to the matching uncompressed student baseline for the same family and scale, which makes the individual techniques and the multi-stage combinations directly comparable at a summary level.

Table 6: Object base compression summary.

Technique	**Δ mAP$_{50–95}$**	**Lat. ↓**
Hardware acceleration	-0.3%	31.3%
Quantisation	-4.9%	56.6%
FP16	-1.6%	53.8%
INT8	-8.2%	59.4%
Pruning	-2.3%	15.4%
90% pruning	-1.6%	13.5%
80% pruning	-2.4%	15.6%
70% pruning	-4.6%	22.0%
Distillation	-1.3%	0.1%
Small jump	-1.1%	-0.5%
Medium jump	-0.7%	0.0%
Large jump	-2.1%	0.9%

Note: Distillation jump size is grouped by teacher-size gap: small = next size up, medium = two sizes up, and large = three or more sizes up.

Table 7: Object combined compression summary.

Combination	**Δ mAP$_{50–95}$**	**Lat. ↓**
Distilled	-4.4%	36.1%
Distilled + Accelerated	-1.5%	23.4%
Distilled + FP16	-2.8%	42.0%
Distilled + INT8	-8.9%	42.9%
Pruned	-5.8%	55.9%
Pruned + Accelerated	-2.8%	40.5%
Pruned + FP16	-4.4%	61.6%
Pruned + INT8	-10.1%	65.6%
Distilled + Pruned	-3.9%	26.6%
Distilled + Pruned	-1.6%	8.3%
Distilled + Pruned + Accel.	-2.2%	23.1%
Distilled + Pruned + FP16	-3.3%	37.7%
Distilled + Pruned + INT8	-8.6%	37.3%

Note: Rows ending with "Accelerated" or "Accel." refer to TensorRT engine deployment without FP16 or INT8 quantisation. FP16 and INT8 rows are also accelerated deployments, but include reduced-precision quantisation as an additional step.

Table 8: Pose base compression summary.

Technique	Δ $mAP_{50–95}$	Lat. ↓
Hardware acceleration	-0.2%	29.9%
Quantisation	-6.5%	59.6%
FP16	-1.3%	56.0%
INT8	-11.8%	63.2%
Pruning	-24.6%	16.3%
90% pruning	-23.7%	15.2%
80% pruning	-24.5%	16.9%
70% pruning	-26.3%	17.3%
Distillation	-4.7%	0.6%
Small jump	-4.8%	0.2%
Medium jump	-6.3%	0.6%
Large jump	-3.0%	1.2%

Note: Distillation jump size is grouped by teacher-size gap: small = next size up, medium = two sizes up, and large = three or more sizes up.

Table 9: Pose combined compression summary.

Combination	Δ $mAP_{50–95}$	Lat. ↓
Distilled	-12.6%	39.7%
Distilled + Accelerated	-4.8%	24.3%
Distilled + FP16	-4.7%	44.9%
Distilled + INT8	-28.2%	49.9%
Pruned	-30.8%	50.5%
Pruned + Accelerated	-24.5%	39.1%
Pruned + FP16	-24.5%	54.4%
Pruned + INT8	-43.3%	58.1%
Distilled + Pruned	-9.7%	33.6%
Distilled + Pruned	-3.4%	11.3%
Distilled + Pruned + Accel.	-3.6%	33.0%
Distilled + Pruned + FP16	-3.7%	43.5%
Distilled + Pruned + INT8	-28.2%	46.7%

Note: Rows ending with "Accelerated" or "Accel." refer to TensorRT engine deployment without FP16 or INT8 quantisation. FP16 and INT8 rows are also accelerated deployments, but include reduced-precision quantisation as an additional step.

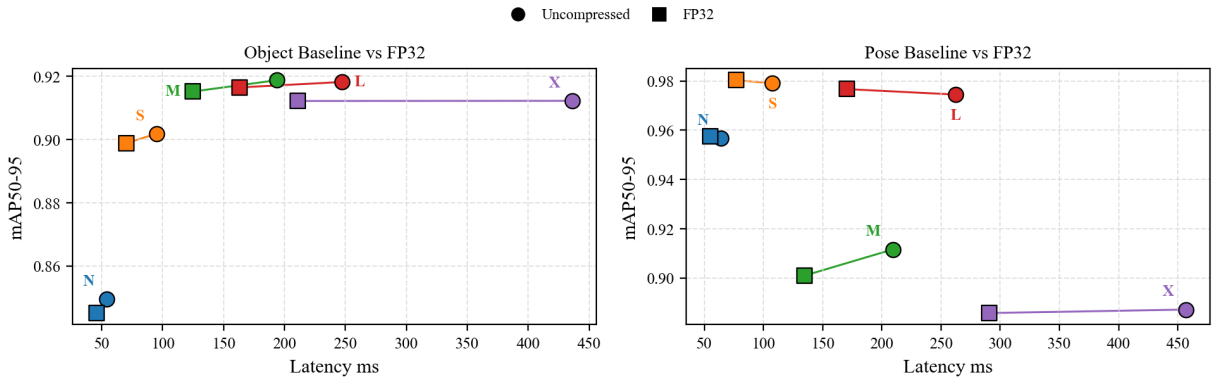
At this higher-level view, the object models benefit most strongly from quantisation in the base-technique table and from pruning–quantisation in the combinations table, since these options produce the largest latency reductions while keeping the mean $mAP_{50–95}$ drop within a moderate range. The pose models are much more sensitive, especially under pruning and INT8-based combinations, where the efficiency gains remain large but the average $mAP_{50–95}$ loss becomes substantially higher. Distillation on its own changes latency

very little in either task, so its main role is not as a standalone acceleration method, but as a supporting step inside broader compression paths.

7.3.1 Hardware Acceleration

Figure 9 presents the average mAP50-95 versus latency relationship for the uncompressed and FP32-accelerated YOLO object and pose model families. By averaging the YOLO11 and YOLO26 variants at each model scale, the figure provides a higher-level view of how deployment through TensorRT-based hardware acceleration affects the overall accuracy–latency trade-off. This allows the effect of acceleration to be separated from architectural differences between individual models, making the general performance shift across scale easier to interpret.

Figure 9: Hardware acceleration trade-off.

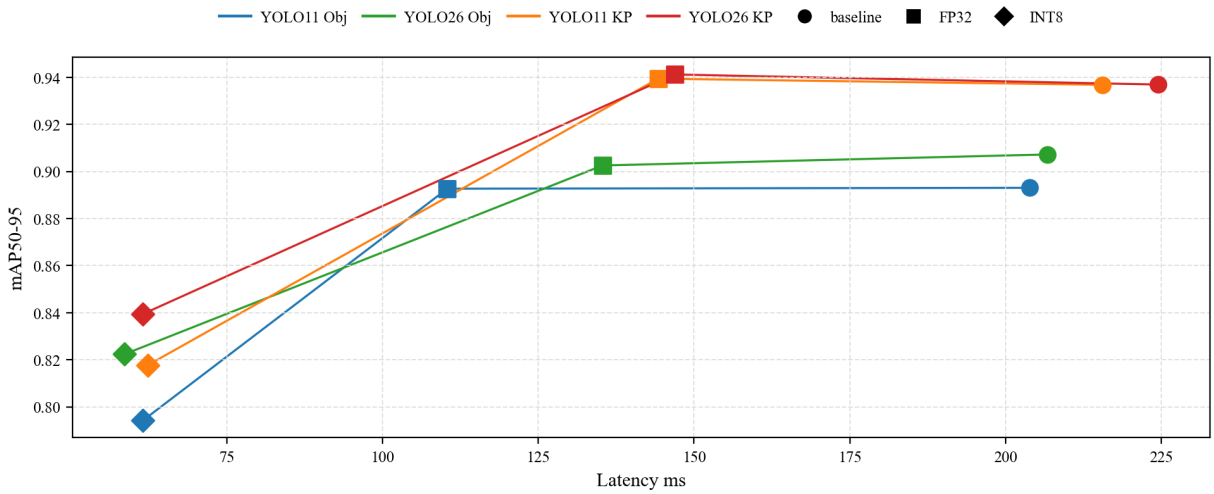


The results show that hardware acceleration provides a substantial reduction in latency across all model scales while preserving almost identical mAP50-95 performance. In each family-task group, the FP32-accelerated points move consistently leftward relative to their uncompressed counterparts, indicating faster inference with negligible accuracy loss. This suggests that deployment optimisation alone provides a strong baseline improvement before more aggressive compression techniques are considered. It is also notable that the relative ordering of the models is preserved after acceleration, meaning that larger models remain more accurate but slower, while smaller models remain faster but less accurate. Overall, hardware acceleration improves deployability without materially altering the underlying accuracy hierarchy.

7.3.2 Quantisation

Figure 10 shows the average mAP50-95 versus latency trade-off for baseline, FP32, and INT8 deployment stages across the YOLO11 and YOLO26 object and pose families. Each line traces the progression of a family-task group through increasingly aggressive deployment optimisation, allowing the effect of quantisation on both inference speed and accuracy retention to be examined in a structured way. This makes it possible to compare whether object and pose models respond similarly to reduced numerical precision.

Figure 10: Quantisation trade-off.

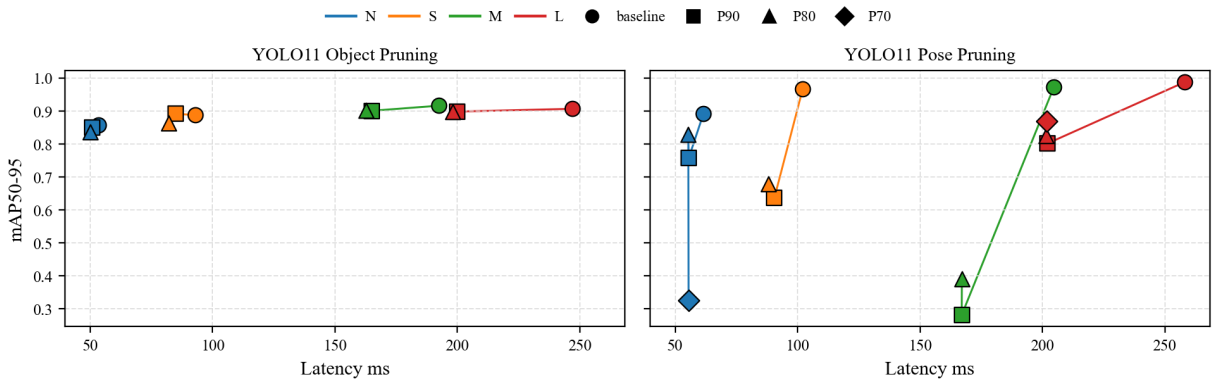


The figure shows that quantisation produces clear latency improvements, with INT8 giving the largest speedup in every family-task group. However, the associated accuracy cost is much more pronounced than in the hardware acceleration case. FP32 acceleration preserves accuracy almost exactly, whereas the move to INT8 shifts the points further left but also noticeably downward. This effect is particularly clear for the object models, where latency is reduced substantially but mAP50-95 falls by a visible margin. The pose models also experience a drop, though they remain at relatively high absolute accuracy. Overall, the results indicate that quantisation is effective for improving runtime efficiency, but that the gain is achieved through a more explicit accuracy–latency trade-off than hardware acceleration alone.

7.3.3 Pruning

Figure 11 illustrates the mAP50-95 versus latency relationship for YOLO11 object and pose models under increasing pruning strength. Separate N, S, M, and L lines are shown for the baseline, P90, P80, and P70 checkpoints, allowing the effect of structured pruning to be examined both across pruning intensity and across model scale. This presentation makes it possible to assess whether pruning shifts the models toward a more favourable operating point or instead introduces excessive accuracy degradation.

Figure 11: Pruning trade-off.

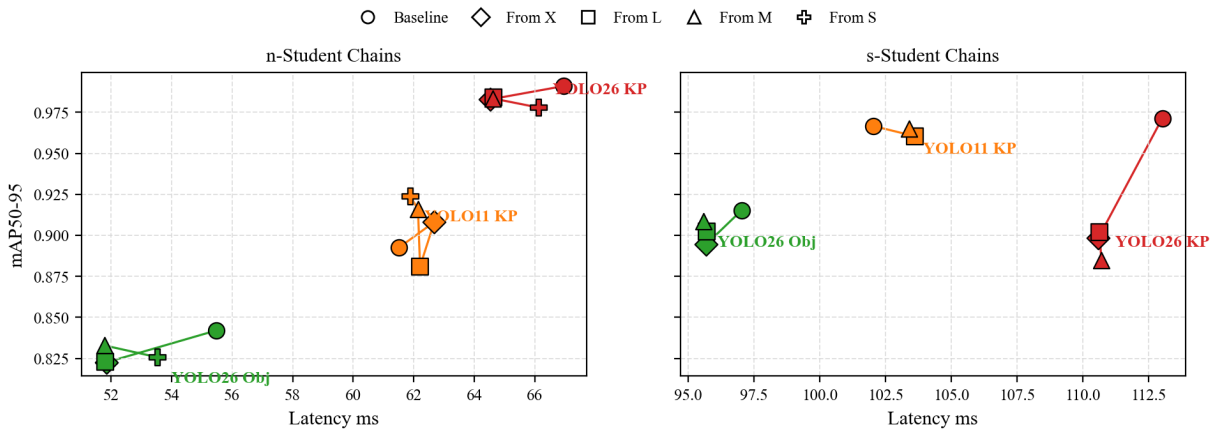


The object detection results show that pruning generally reduces latency while causing only moderate reductions in mAP50-95, particularly for the larger model scales. In several cases, the pruned object models remain close to the baseline in accuracy while offering meaningful speed improvements, which suggests that object detection models retain a degree of redundancy that pruning can exploit. In contrast, the pose models are much more sensitive to pruning. While mild pruning can still provide a useful trade-off, heavier pruning leads to severe reductions in mAP50-95 for several scales, especially at the strongest pruning settings. This indicates that the keypoint models are less robust to parameter removal and reach an accuracy floor much more quickly. Overall, the figure suggests that pruning is a more reliable compression method for object detection than for pitch keypoint estimation.

7.3.4 Knowledge Distillation

Figure 12 presents the effect of knowledge distillation on YOLO11s object and pose models by comparing the baseline student against students distilled from larger teacher models. The figure shows whether transferring knowledge from progressively larger teachers produces measurable improvements in the student accuracy–latency trade-off, while keeping the underlying student architecture fixed. This makes it possible to isolate the contribution of distillation itself rather than conflating it with changes in model size.

Figure 12: Knowledge distillation trade-off.

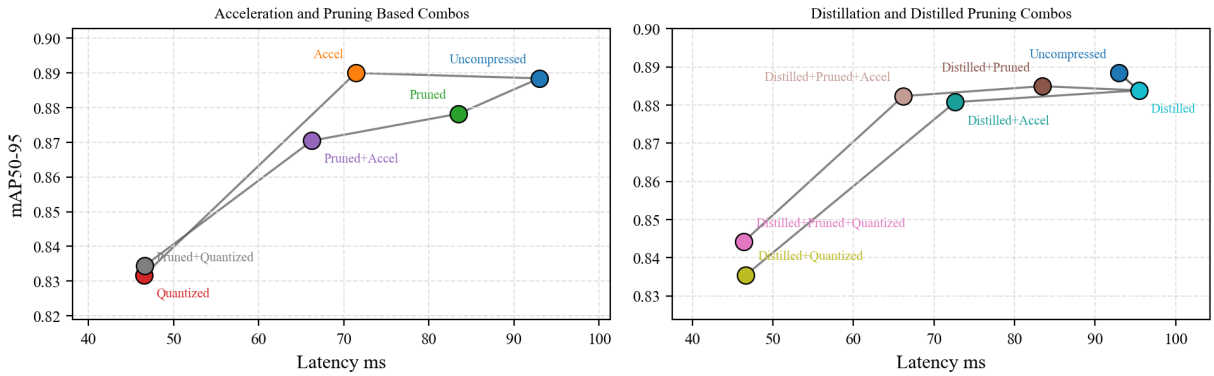


The results show that distillation produces modest but consistent accuracy improvements while leaving latency almost unchanged. For the object student, the distilled variants cluster very closely in latency yet slightly outperform the baseline in mAP50-95, indicating that the student benefits from teacher guidance without incurring a meaningful runtime cost. A similar pattern is visible for the keypoint student, where the distilled models remain close to the baseline latency while achieving comparable or slightly improved accuracy. The gains are not dramatic, but they are important because they move the student upward rather than forcing the usual downward shift associated with compression. Overall, the figure suggests that knowledge distillation is an accuracy-preserving compression support technique rather than a direct latency-reduction method, and is most valuable when used to strengthen smaller deployable models.

7.3.5 Combinations of Different Techniques

Figure 13 brings the different compression strategies together by tracing the mAP50-95 versus latency paths of YOLO11s object models under acceleration, pruning, quantisation, distillation, and their combined variants. Rather than treating each technique in isolation, the figure shows how successive optimisation steps reshape the overall accuracy–latency frontier. This provides a compact way to compare which sequences of transformations produce the most favourable deployment trade-offs.

Figure 13: Combined compression trade-off.



The combined results show that no single path dominates in all respects, and that the best choice depends on whether latency reduction or accuracy retention is prioritised. Acceleration alone provides a strong first improvement, while quantisation gives the largest latency reduction but also the largest drop in mAP50-95. Pruning tends to occupy the middle ground, offering useful speed gains with smaller accuracy loss than quantisation. Distillation-based paths are especially notable because they help preserve or recover accuracy, making the combined distilled-pruned or distilled-accelerated variants more balanced overall. The figure therefore reinforces the broader pattern seen throughout this section: the most effective deployment candidates are usually not the most aggressively compressed models, but the intermediate configurations that balance accuracy preservation with meaningful runtime improvement.

7.4 Hardware Analysis

7.4.1 Latency

Table 10 summarises how each compression stage changes end-to-end inference latency across the object and pose models. To keep the comparison compact, the values average the paired YOLO11 and YOLO26 results within each size class, while still separating the object and pose tasks. This makes it easier to see whether the same optimisation strategy produces similar runtime benefits across both detection and keypoint estimation.

Table 10: Average latency reduction by stage.

	Object					**Pose**				
Stage	**N**	**S**	**M**	**L**	**X**	**N**	**S**	**M**	**L**	**X**
Uncompressed	54.37	95.00	193.81	247.34	436.18	64.23	107.53	209.28	262.16	456.87
Accelerated	15.7%	26.1%	35.7%	34.1%	51.8%	14.3%	28.2%	35.7%	35.1%	36.4%
Distilled	15.8%	25.6%	36.4%	34.6%	–	15.5%	28.7%	36.3%	36.2%	–
Pruned	13.0%	31.8%	43.8%	46.8%	–	27.5%	33.8%	46.1%	47.9%	–
Quantized	21.1%	52.5%	71.8%	74.2%	78.5%	32.7%	57.1%	72.3%	75.0%	78.7%

The latency results show that quantisation delivers the largest and most consistent speed-up for both tasks, with reductions rising from around one fifth at the smallest object models to almost 79% for the X variants. Acceleration also produces a strong baseline improvement, particularly for the larger models where the absolute runtime cost is highest, while pruning offers a more moderate middle ground on the sizes where paired runs are available. Distillation improves latency only modestly by comparison, which reinforces the earlier observation that its main contribution is preserving accuracy rather than directly driving runtime reductions.

7.4.2 GPU Usage

Table 11 reports the corresponding change in GPU utilisation after each compression stage. Because the same averaging scheme is used here, the table allows the hardware load of object and pose inference to be compared directly across model scale, while also showing whether reductions in latency are accompanied by lower pressure on the GPU.

Table 11: Average GPU reduction by stage.

	Object					Pose				
Stage	**N**	**S**	**M**	**L**	**X**	**N**	**S**	**M**	**L**	**X**
Uncompressed	63.23	76.81	87.44	89.50	92.61	67.03	78.76	87.71	91.62	93.04
Accelerated	17.7%	21.2%	13.9%	10.2%	5.1%	17.7%	19.8%	14.2%	11.9%	9.8%
Distilled	8.5%	10.9%	6.4%	5.0%	–	19.8%	20.3%	12.6%	11.9%	–
Pruned	14.5%	12.1%	11.3%	7.6%	–	23.9%	23.4%	17.0%	17.3%	–
Quantized	20.8%	32.5%	33.9%	33.5%	24.1%	25.3%	40.7%	36.7%	33.6%	25.9%

The GPU results broadly follow the latency trends, but with smaller absolute gains for most techniques. Quantisation again gives the largest reductions, especially for the S, M, and L variants, while hardware acceleration offers reliable but more moderate savings. Distillation and pruning reduce GPU usage in several cases, but the improvements are clearly less dramatic than those seen for latency, suggesting that these techniques reshape the compute profile only partially rather than fundamentally lowering accelerator demand.

7.4.3 CPU Usage

Table 12 shifts the focus from accelerator load to host-side overhead. This comparison is particularly useful because some optimisations that improve inference speed on the GPU can still increase CPU involvement through scheduling, memory movement, or engine management, so the table helps expose trade-offs that are not visible from latency alone.

Table 12: Average CPU reduction by stage.

	Object					Pose				
Stage	**N**	**S**	**M**	**L**	**X**	**N**	**S**	**M**	**L**	**X**
Uncompressed	13.34	8.69	6.16	5.62	5.46	12.49	8.39	5.91	6.44	6.01
Accelerated	29.0%	14.1%	22.3%	26.2%	51.6%	28.6%	14.1%	22.2%	38.3%	59.5%
Distilled	14.3%	7.3%	11.8%	10.4%	–	28.8%	14.2%	23.1%	38.5%	–
Pruned	10.6%	0.9%	0.5%	-1.2%	–	21.5%	8.0%	5.1%	23.7%	–
Quantized	25.8%	-9.8%	-34.5%	-36.2%	-12.7%	20.6%	-18.6%	-44.8%	-22.8%	-0.3%

The CPU results are more mixed than the GPU results. Hardware acceleration gives the clearest CPU benefit overall, especially for the X models, whereas pruning yields only small

gains and in some object cases is close to neutral. Most notably, quantisation frequently increases CPU usage for the medium, large, and extra-large models, producing negative reductions, which suggests that part of the optimisation cost is being shifted away from the accelerator and onto the CPU. Distillation sits between these extremes, with moderate CPU savings for the available N–L sizes but no evidence that it is a dominant hardware-efficiency technique by itself.

7.4.4 Power Draw

Table 13 shows how the same transformations affect total power draw. Since deployment on embedded hardware is constrained not only by raw throughput but also by energy availability and thermal headroom, these measurements provide a more deployment-oriented view of which techniques meaningfully reduce system-wide operating cost.

Table 13: Average power reduction by stage.

	Object					**Pose**				
Stage	**N**	**S**	**M**	**L**	**X**	**N**	**S**	**M**	**L**	**X**
Uncompressed	12.77	14.99	16.53	17.00	17.62	13.30	15.07	16.52	17.16	17.65
Accelerated	18.2%	10.9%	3.5%	3.6%	-1.2%	45.9%	31.9%	17.5%	16.9%	-0.7%
Distilled	19.1%	13.0%	2.5%	3.8%	–	-0.7%	-0.7%	0.3%	0.0%	–
Pruned	20.7%	15.6%	6.1%	6.5%	–	2.8%	1.2%	1.2%	2.1%	–
Quantized	53.1%	53.1%	45.4%	36.4%	23.7%	52.2%	50.8%	44.0%	34.8%	24.0%

The power results indicate that quantisation is the strongest method for reducing energy demand, with savings of roughly 24% to 53% across the reported sizes. Hardware acceleration also lowers power substantially for the smaller pose models, but its effect weakens for larger object models and is almost neutral for the X variants. By contrast, distillation and pruning have relatively small influence on power draw, which implies that their main value lies elsewhere, such as accuracy retention or balanced latency reduction, rather than direct energy savings.

7.4.5 Temperature

Table 14 extends the hardware analysis to thermal behaviour. Temperature is an important deployment metric because sustained improvements in latency or power are only truly useful if they also reduce heat generation enough to avoid thermal throttling and maintain stable operation over long runs.

Table 14: Average temperature reduction by stage.

	Object					**Pose**				
Stage	**N**	**S**	**M**	**L**	**X**	**N**	**S**	**M**	**L**	**X**
Uncompressed	58.79	60.43	63.14	63.09	64.64	58.16	60.97	63.16	64.14	64.76
Accelerated	5.7%	1.6%	-0.3%	-1.1%	-1.5%	15.1%	11.0%	6.1%	5.6%	-0.7%
Distilled	6.8%	1.2%	-0.5%	-1.2%	–	-0.6%	-0.3%	1.5%	0.2%	–
Pruned	6.8%	3.1%	1.3%	-0.6%	–	0.6%	1.1%	1.5%	1.1%	–
Quantized	19.1%	16.2%	13.7%	9.8%	6.7%	16.4%	18.0%	15.3%	11.9%	8.3%

The temperature results again favour quantisation, which produces the largest and most uniform reductions across both object and pose models. Hardware acceleration helps noticeably for the smaller and mid-sized pose models, but has only limited or even slightly negative impact for the larger object variants, showing that faster execution does not automatically translate into cooler operation. Distillation and pruning have the weakest and least consistent thermal effects, so while they remain useful from an accuracy–efficiency perspective, they are not the primary tools for improving thermal behaviour.

7.5 Full-Stack System

7.6 Full-Stack System Implementation

7.6.1 Full-Stack System Performance

With individual model compression analysis complete, the Pareto frontier analysis discussed earlier can be implemented. This was used to identify models that offered the strongest balance between analytical performance and hardware efficiency [19]. For object detection, the frontier shortlist consisted of four compressed YOLO26 variants: **26l**

(FP16), **26s (Baseline)**, **26m (80%, FP16)**, and **26s (FP16)** [See Table 15 for an explanation of this labelling]. For keypoint estimation, the frontier was composed entirely of compressed 26n-pose variants: **26n (FP16)**, **26n (80%, FP16)**, **26n (KD-26x, 80%, FP16)**, and **26n (KD-26m, 80%, INT8)**.

These shortlisted models were then evaluated in the proposed full-stack pipeline to analyse higher-level statistic accuracy and compared to the prior SoccerNet system, which achieved a pass $\text{mAP}_{1\text{–}5}$ of 71.2% [3], with $\text{mAP}_{1\text{–}5}$ defined earlier. Each model combination was run on a continuous 45 minute football video within the proposed system. Table 15 reports the five strongest full-stack configurations, with "$\Delta\text{mAP}_{1\text{–}5}$ vs. SoccerNet" representing the percent difference in $\text{mAP}_{1\text{–}5}$ against the SoccerNet system. The table also reports the difference between predicted and ground-truth possession, together with runtime hardware statistics.

Table 15: Top five full-stack combinations.

Object Model	**Keypoint Model**	**Pass $\text{mAP}_{1\text{–}5}$**	**ΔPos vs. GT**	**FPS**	**GPU (%)**
26m (80%, FP16)	**26n (80%, FP16)**	**0.697**	**12.22%**	**6.51**	**39.61**
26l (FP16)	26n (FP16)	0.687	1.11%	5.57	51.50
26l (FP16)	26n (KD-26m, 80%, INT8)	0.655	7.71%	5.75	53.39
26m (80%, FP16)	26n (KD-26x, 80%, FP16)	0.583	10.83%	6.49	44.27
26s (Uncompressed)	26n (KD-26m, 80%, INT8)	0.574	7.54%	6.18	47.64

Note: Model labels use the format *Architecture (Compression Settings)*. Here, 80% denotes 80% structured pruning, KD-26m and KD-26x denote knowledge distillation from 26m and 26x teacher models respectively, and FP16 and INT8 denote quantisation. ΔPos vs. GT gives the absolute difference in dominant-team possession share relative to ground truth.

As shown in Table 15, all five configurations remained within **15%** of the SoccerNet $\text{mAP}_{1\text{–}5}$ baseline, indicating that the compressed models and optimised full-stack system can achieve performance close to that of non-real-time systems. This is encouraging, as it shows that competitive pass-detection performance can be retained while operating close to real-time on the Jetson Orin Nano. The strongest overall configuration achieved the smallest gap to the baseline at just **5.5%**, with other model combinations also remaining reasonably close to the baseline.

Table 16 provides the corresponding average hardware profile across all 16 evaluated combinations in the same rank-02 replay batch.

Table 16: Average full-stack hardware profile.

Summary	Latency (ms)	FPS	Power (W)	Temp (°C)	GPU (%)
Mean across 16 combos	75.89	13.17	3.83	52.34	47.35

Figures 14 and 15 complement these table summaries by plotting pass mAP_{1-5} against the main runtime and hardware metrics for all 16 rank-02 full-stack combinations. This makes it easier to see whether stronger analytical performance is tied to substantially higher latency, GPU load, power draw, or temperature.

Figure 14: Pass mAP_{1-5} versus runtime metrics for rank-02 full-stack combinations.

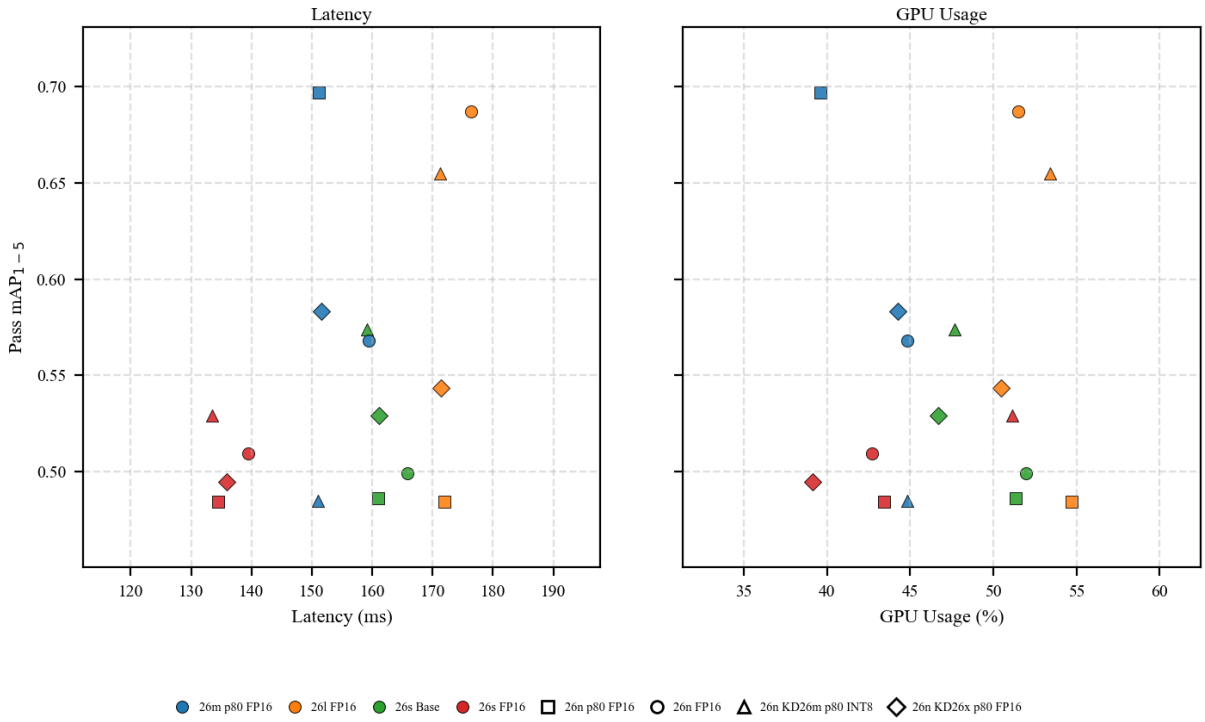
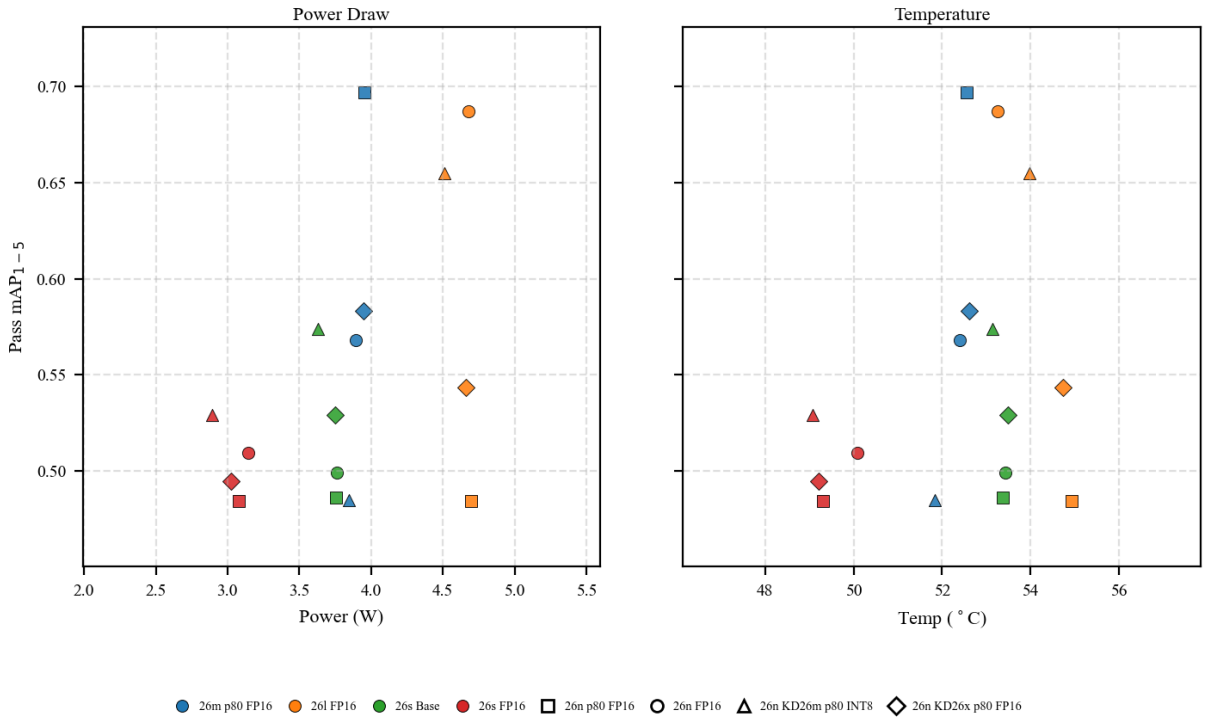


Figure 15: Pass mAP_{1-5} versus thermal and power metrics for rank-02 full-stack combinations.



Taken together, these figures show that the strongest pass-detection results do not require the highest hardware load within the tested full-stack shortlist. The higher-performing combinations cluster within a fairly narrow operating band, with only moderate variation in GPU usage, power draw, and temperature, while the latency plot shows that the best analytical scores are not confined to the slowest configurations. This reinforces the conclusion from Table 16: the system operates comfortably within the limits of the NVIDIA Jetson Orin Nano Dev Kit, with power draw far below the 25 W budget, temperature well under the 85°C limit, and GPU utilisation below 55% in every case. The systems reached effective throughputs of roughly 11–13 FPS, near-real-time. Although this still falls short of strictly real-time (generally considered 20–25 FPS in broadcast video), the remaining gap is small, suggesting that full real-time deployment could likely be achieved through further system-level optimisation, rather than substantial additional model compression.

8 Discussions and Conclusions

8.1 Discussion of Results

This thesis set out to examine whether computer vision models for soccer analysis could be trained, compressed, and deployed effectively on embedded hardware without sacrificing too much analytical performance. The results indicate that this is achievable, but only when compression is applied selectively and interpreted in the context of the target task. Across both object detection and pitch keypoint localisation, the most useful deployment candidates were not the most aggressively compressed models, but the configurations that preserved enough accuracy while still reducing latency, power draw, and thermal load to make edge deployment practical.

A clear theme throughout the results is that hardware acceleration should be treated as the baseline optimisation step rather than as an optional enhancement. Figure 9 showed that TensorRT-based FP32 acceleration reduced latency substantially while preserving almost identical $mAP_{50\text{–}95}$. This is important because it means a considerable portion of the deployment benefit can be obtained before any accuracy-sensitive compression is introduced. In practical terms, the acceleration stage improves real-time viability without changing the relative ranking of the models, so model selection can still be based primarily on the original accuracy–latency balance.

Quantisation emerged as the strongest method for improving embedded efficiency overall. The latency, power, and temperature results consistently showed that INT8 deployment produced the largest reductions in computational cost. However, Figure 10 and the summary tables also showed that these gains came with a more explicit accuracy penalty than hardware acceleration alone. This trade-off is especially important for edge sports analytics, where improved throughput is valuable only if the extracted statistics remain trustworthy. The quantisation results therefore suggest that INT8 deployment is most appropriate when strict runtime or energy limits dominate, whereas FP32-accelerated models remain preferable when analytical fidelity is the primary concern.

One of the most important findings is that compression tolerance depends strongly on the

vision task. The object detection models were consistently more robust than the pitch keypoint models under pruning and combined compression paths. Figure 11 suggests that the object models retained enough representational redundancy for structured pruning to reduce latency without catastrophic accuracy loss, particularly at larger scales. In contrast, the pose models degraded much more quickly as pruning strength increased. A plausible interpretation is that pitch keypoint localisation depends on finer spatial structure and more fragile feature relationships than player and ball detection, making it less tolerant to parameter removal and reduced numerical precision. This task-dependent sensitivity means that a single compression recipe cannot be assumed to generalise across all stages of a sports analytics pipeline.

Knowledge distillation occupied a different role from the other techniques. As shown in Figure 12, distillation did not materially change latency on its own, but it often helped smaller models retain or slightly improve analytical performance. This makes distillation less valuable as a direct deployment optimisation and more valuable as a supporting mechanism that stabilises compressed students. The combined-path analysis in Figure 13 reinforces this interpretation, since the strongest deployment candidates were typically balanced intermediate configurations in which acceleration and moderate compression were paired with accuracy-preserving steps such as distillation. In other words, the results support a staged view of optimisation: first accelerate, then compress cautiously, and use distillation where accuracy recovery is needed.

The hardware measurements also add an important practical layer to the discussion. Improvements in latency did not always translate proportionally to reductions in GPU or CPU usage, and quantisation in particular sometimes increased CPU usage for larger models. This suggests that some deployment benefits are achieved by shifting the system bottleneck rather than eliminating it entirely. For an embedded platform, that distinction matters. A model that is faster but transfers more overhead to the host may still be useful, but it changes the design constraints for the rest of the pipeline. The power and temperature results therefore strengthen the case for quantisation as a deployment tool, since they show that its value is not limited to faster inference alone, but extends to lower sustained

operating cost and improved thermal behaviour.

The full-stack replay experiments show that these model-level findings also translate into system-level usefulness. The best full-stack configuration achieved a Pass $mAP_{1\text{–}5}$ of 0.697, 13.02 FPS, and 39.61% GPU utilisation, demonstrating that embedded soccer analytics can be carried out on the Jetson Orin Nano without saturating the device. At the same time, these results also show that the original real-time target of roughly 20 FPS was not fully reached by the final end-to-end pipeline. This is an important nuance. The thesis still demonstrates practical viability, especially for statistics such as possession and pass analysis that evolve over multiple frames, but it also shows that model compression alone is not sufficient to close the remaining gap to higher-throughput real-time operation. System-level optimisations in scheduling, detection interval control, and asynchronous processing remain necessary.

8.2 Limitations

Several limitations should be considered when interpreting the results. First, the benchmarking was carried out on a single target platform, the Jetson Orin Nano Dev Kit, so the absolute latency and hardware measurements should not be assumed to transfer directly to other embedded devices. Second, the model training and evaluation were based on a selected set of football broadcast datasets, which means that generalisation to lower-quality footage, unusual camera angles, severe occlusion, or non-broadcast match settings remains uncertain. Third, the full-stack analysis focused on a constrained set of match statistics and replay conditions rather than a fully production-ready analytics system. Finally, while a broad set of model artefacts was evaluated, the study necessarily sampled only part of the wider compression design space, especially for combined techniques and longer-duration deployment behaviour.

Despite these limitations, the thesis makes a useful contribution by linking model compression analysis to an operational soccer analytics pipeline rather than treating benchmark performance in isolation. The results show not only which techniques improve latency, but which ones remain meaningful once power, thermal behaviour, and end-to-end analytical

usefulness are considered together.

8.3 Conclusion

This thesis investigated how computer vision models for soccer analysis can be trained, compressed, and deployed efficiently on embedded hardware. Across 488 trained and benchmarked model artefacts, the results showed that deployment-focused optimisation can substantially improve the practicality of edge-based sports analytics while preserving useful analytical performance. Hardware acceleration provided the safest and most broadly effective first step, quantisation delivered the strongest reductions in latency and operating cost, pruning proved more suitable for object detection than for pitch keypoint localisation, and knowledge distillation was most valuable as an accuracy-preserving support technique rather than as a direct speed-up method.

The work also demonstrated that model-level improvements can be carried through to a full-stack soccer analysis system for detection, localisation, possession estimation, and pass detection on the Jetson Orin Nano. Although the final system did not fully meet the most ambitious throughput target, it achieved a practically useful operating point at 13.02 FPS with moderate GPU utilisation and meaningful analytical output. Overall, the thesis shows that real-time or near-real-time soccer analytics on constrained edge hardware is achievable, but that the best deployment candidates are balanced compressed models rather than the most aggressively optimised ones. This provides a clear foundation for future work on system-level optimisation and broader match understanding, and supports the wider goal of low-cost, on-site analytics for teams, broadcasters, and sports science applications.

9 Impact of Research

This research has the potential to deliver impact by making soccer analytics more accessible, more energy-efficient, and less dependent on expensive cloud-based or specialist commercial infrastructure. While the work presented in this thesis is still at a prototype

and benchmarking stage, it demonstrates that useful computer-vision-driven match analysis can be performed on an embedded edge device, which creates a realistic path toward lower-cost deployment in coaching, analysis, and broadcasting settings.

9.1 Potential Beneficiaries and Pathways to Impact

The most immediate beneficiaries of this work are researchers and engineers working on embedded computer vision, since the thesis provides a structured comparison of deployment-focused optimisation methods across object detection, pitch keypoint localisation, and full-stack soccer analysis. The results also have potential value for sports science groups, university teams, amateur and semi-professional clubs, and smaller broadcasters, all of whom may lack the budget required for commercial analytics providers or high-end cloud infrastructure.

In the short term, the main impact of the project is methodological. The benchmarking framework, the compression analysis, and the integrated pipeline provide evidence for which optimisation strategies are most effective when the goal is to balance accuracy with real-time or near-real-time operation on constrained hardware. This can inform future academic work and help guide deployment decisions for similar sports analytics systems.

In the medium term, this research could support the development of low-cost tools for match review, player development, and local broadcasting. For example, a refined version of the system could help coaches and analysts generate statistics such as possession trends, pass patterns, and player movement indicators without the need for large-scale cloud processing. This could widen access to data-driven decision-making beyond elite clubs and make analytical tools available to organisations with more limited resources.

In the longer term, if the methods were extended into a more robust production system, the work could contribute economically by supporting innovation in sports technology products and services. This includes possible applications in affordable analytics platforms, edge-deployed performance tools, and broadcast support systems. The impact would not arise from the prototype alone, but from the adoption and refinement of its outputs into

deployable products, workflows, or operational standards.

9.2 Ethical Considerations and Risk Mitigation

Although the project is motivated by performance analysis rather than surveillance, it still raises important ethical considerations. The first concerns the use of player image data in training and evaluation datasets. In this work, the datasets were drawn from publicly available or appropriately shared football video sources used for non-commercial research purposes. In addition, the system does not perform facial recognition, biometric identification, or any attempt to infer personal identity. Players are represented only through bounding boxes, keypoints, and team-level analytical outputs, which reduces the privacy sensitivity of the system compared with person-identification technologies.

However, some risks remain. A system of this type could be used inappropriately for unauthorised monitoring of athletes, or its outputs could be over-interpreted in settings where model predictions are uncertain. Dataset bias is also a concern, since models trained primarily on professional broadcast footage may not generalise equally well to lower-quality video, different lighting conditions, youth matches, or non-broadcast camera angles. If such a system were used without caution, inaccurate detections or biased model behaviour could influence tactical decisions, player evaluation, or analyst confidence in misleading ways.

These risks can be mitigated in several ways. First, the intended use of the system should remain limited to performance analysis rather than identity-based tracking. Second, model outputs should be treated as decision-support tools rather than fully autonomous judgements, with human review maintained for important tactical or evaluative decisions. Third, deployment should include transparent reporting of uncertainty, known failure cases, and data limitations. Finally, any future expansion toward commercial or operational use should ensure that dataset licensing, consent requirements, and governance procedures are clearly respected.

9.3 Sustainability Considerations and Sustainable Development Goals

This project is also relevant to sustainability because it focuses on computational efficiency rather than maximising performance through ever-larger models and unrestricted compute usage. Deep learning systems can carry a significant environmental cost when they rely on repeated training cycles or cloud-scale inference without regard for efficiency [4]. By evaluating hardware acceleration, quantisation, pruning, and knowledge distillation for embedded deployment, this thesis instead promotes an approach in which useful analytical performance is achieved with lower power draw, reduced thermal load, and less dependence on continuous data transfer to centralised servers.

The work aligns most directly with United Nations Sustainable Development Goal 9, *Industry, Innovation and Infrastructure*, by contributing toward practical and affordable digital infrastructure for applied sports analytics. It also aligns with Sustainable Development Goal 12, *Responsible Consumption and Production*, by emphasising efficient model design and resource-aware deployment rather than unnecessary compute expansion. In addition, it has relevance to Sustainable Development Goal 13, *Climate Action*, since edge-based inference and model compression can reduce the energy overhead associated with cloud processing and prolonged high-power operation.

At the same time, sustainability is not automatically guaranteed simply because inference is performed on-device. Training many model variants can still consume substantial energy, and future development could increase hardware turnover or encourage inefficient experimentation if left unmanaged. For that reason, this project has aimed to limit redundant training runs, reuse model outputs where possible, and prioritise controlled benchmarking. A responsible continuation of this research would continue to favour energy-aware evaluation, selective experimentation, and deployment choices that are justified by clear analytical benefit rather than by marginal performance gains alone.

10 Suggestions for Future Work

The future directions of this project follow directly from the main limitations identified in the discussion. Although the thesis demonstrated that embedded soccer analytics is feasible on the Jetson Orin Nano, the evaluation was limited to a single hardware platform, a selected set of broadcast-style datasets, a constrained set of full-stack statistics, and a partial exploration of the broader compression design space. In addition, the final end-to-end system achieved a practically useful operating point, but did not fully reach the most ambitious real-time throughput target. Future work should therefore not be viewed simply as feature expansion, but as a set of focused research investigations aimed at improving generalisability, deployment reliability, and end-to-end analytical value.

10.1 Extending Full-Stack Match Understanding

One important direction is to extend the system beyond the current set of outputs toward richer match statistics and event understanding. In particular, future work could investigate pass completion percentage, turnovers, ball progression sequences, possession chains, and more detailed player movement summaries. This would make the system more useful to coaches, analysts, and broadcasters, and would help bridge the gap between low-level detections and higher-level football understanding.

However, this extension is not a straightforward implementation task. Reliable pass completion or turnover analysis depends on stable ball detection, robust player tracking, accurate team assignment, and temporally consistent possession reasoning. Errors in any one stage can propagate to the final statistic, especially during occlusion, rapid camera motion, or ambiguous multi-player interactions. Future studies should therefore examine temporal modelling strategies, confidence-aware event rules, and methods for validating event-level outputs against manually annotated sequences. The impact of this work would be significant, since it would determine whether embedded systems of this type can move beyond demonstration-level analytics and provide outputs that are useful in realistic tactical or broadcast settings.

10.2 System-Level Optimisation for Real-Time Deployment

A second major area for further investigation is end-to-end system optimisation. The results showed that model compression alone was not sufficient to close the remaining gap to the target real-time operating point, which suggests that pipeline-level design is now a central research problem. Future work should evaluate asynchronous execution between inference and post-processing, adaptive object-detection intervals, less frequent team-update scheduling, and hybrid approaches in which some pitch or geometric operations are replaced with lighter classical methods [25]. Another promising direction would be to study dynamic model selection, where smaller or faster models are used under high-load conditions and more accurate models are used when the scene is less demanding.

The main difficulty here is that throughput improvements may come at the cost of stale detections, delayed state updates, or unstable inter-stage synchronisation. A faster system is not necessarily a better system if the resulting statistics become temporally inconsistent or analytically unreliable. Future research should therefore measure not only FPS, but also the statistical effect of scheduling decisions on possession estimation, pass detection, and tracking continuity. This line of work is important because it is likely to determine whether the prototype can become a genuinely real-time embedded application rather than a near-real-time research demonstrator.

10.3 Advanced Compression and Training Strategies

The compression study in this thesis established broad trends, but several important directions remain open. Future work should investigate quantization-aware training in addition to post-training quantization, since the results suggest that INT8 deployment offers strong efficiency gains but can introduce substantial accuracy loss. It would also be valuable to compare structured and unstructured pruning more systematically, explore sparsity-aware retraining, and test a wider range of teacher-student pairings for knowledge distillation. The preliminary input-resolution experiments could also be extended by training and evaluating models under controlled resolution mismatch, to determine whether low-cost training or inference configurations can retain acceptable robustness.

These studies are likely to be challenging because compression behaviour is highly task-dependent. The results already showed that pitch keypoint models are much more sensitive than object detectors to aggressive pruning and reduced numerical precision, so future work will need careful experimental design to avoid drawing conclusions from unstable or unfair comparisons. There is also a practical cost in terms of training time, hardware usage, and calibration effort when many compression variants are evaluated. Despite these difficulties, the impact of such work would be substantial: a better understanding of compression-aware training could improve the accuracy–efficiency frontier of embedded sports analytics and produce deployment candidates that are both faster and more trustworthy.

10.4 Robustness, Generalisation, and Cross-Platform Validation

Another key limitation of the present work is that the evaluation was carried out under relatively controlled conditions. Future research should therefore examine how well the trained and compressed models generalise to lower-quality footage, non-broadcast camera placements, different lighting conditions, amateur or youth-level matches, and more diverse pitch environments. In parallel, the full-stack system should be tested over longer-duration runs to evaluate thermal stability, sustained throughput, and failure modes under extended operation. Cross-platform benchmarking on additional edge devices would also be valuable, since deployment trade-offs may differ considerably across CPU, GPU, and memory configurations.

The main challenge in this direction is that robust evaluation requires new annotated data, careful licensing and collection procedures, and access to multiple hardware platforms. Domain shift may also make it difficult to determine whether performance degradation is caused primarily by the model, the compression method, or the broader system assumptions. Even so, this is one of the most important areas for future work, because it determines whether the conclusions of this thesis remain valid outside the specific datasets and hardware used here. Demonstrating broader robustness would strengthen both the scientific contribution and the practical impact of the work, especially for lower-budget organisations that may operate in less controlled recording conditions than professional

broadcast environments.

10.5 Research Value of Continued Development

Taken together, these future directions show that the next stage of this project should focus on transforming a strong prototype into a more general and dependable embedded analytics framework. The importance of carrying out this work lies not only in improving one specific soccer-analysis system, but in advancing the broader understanding of how modern vision pipelines can be compressed, validated, and deployed responsibly on constrained hardware. Further research in these areas could benefit both the sports analytics domain and the wider field of embedded AI, where the same questions of efficiency, robustness, and practical usefulness arise across many real-world applications.

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11 appendices